PROJECT DESIGN PHASE-II

Solution Requirements (Functional & Non-functional)

Date	29 October 2026	Maximum Marks	4 Marks
Team ID	EV-BI-04	Project Name	Electric Vehicle Market Performance Insights Dashboard

Functional Requirements

Following are the functional requirements of the proposed business intelligence solution specifying data access, transformation, analysis, visualization, and reporting capabilities.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority
FR-1	**Data Access & Integration**	• Import EV dataset from Excel/CSV into Tableau environment. • Connect Tableau visual dashboards cleanly with Flask website backend.	High
FR-2	**Data Processing**	• Clean missing values and standardize data formats. • Transform battery specs, vehicle pricing, and historical sales data.	High
FR-3	**Business Analysis**	• Analyze total EV sales performance across years and regions. • Compare major EV manufacturers and vehicle categories. • Analyze battery technologies, pricing insights, and customer ratings.	High
FR-4	**Business Insights & Analytical Outputs**	• Generate interactive Tableau dashboards with dynamic charts and KPIs. • Develop a Story Dashboard to highlight strategic market trends and executive insights.	High
FR-5	**User Interaction & Analysis**	• Provide seamless dashboard navigation. • Enable multi-level filter controls (by manufacturer, category, region). • Allow drill-down analysis into specific EV models and metrics.	Medium
FR-6	**Data Update & Reporting**	• Automatically update visual dashboards whenever underlying dataset changes. • Export analytical reports and visual summaries for business users.	Medium

Non-Functional Requirements

Following are the quality attributes and technical constraints of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Intuitive, clean, and easy-to-use user interface requiring minimal end-user training.
NFR-2	**Security**	Secure storage of project files, application source code, and repository credentials.
NFR-3	**Reliability**	Stable performance; website and embedded components run seamlessly without runtime errors.
NFR-4	**Performance**	Fast dashboard rendering and page response times under standard web network speeds.
NFR-5	**Availability**	High availability; hosted and accessible online 24/7 via Tableau Public & web server.
NFR-6	**Scalability**	Modular architecture allowing future datasets, metrics, and new dashboards to be integrated easily.